

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398890
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Kim; Donghyun et al.

Air conditioner

Abstract

An air conditioner including: a housing communicating with a space in which a window covering is installed, and including a suction port through which air is suctioned from the space and a discharge port through which air is discharged into an indoor area; a heat exchanger configured to exchange heat with the air suctioned through the suction port; a blower fan allowing air to be suctioned through the suction port or allowing air to be discharged through the discharge port; and a guide panel provided at the suction port to guide the window covering to prevent the window covering from covering the suction port.

Inventors: Kim; Donghyun (Suwon-si, KR), Hong; Jinwoo (Suwon-si, KR), Kim; Sungwoo (Suwon-si, KR), Kim; Jinbaek (Suwon-si, KR), Kim; Hyojin (Suwon-si, KR), Sim; Jaehyoung (Suwon-si, KR), Yoon; Seonghyun (Suwon-si, KR), Cho; Sungjune (Suwon-si, KR)

Applicant: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

Family ID: 1000008782437

Assignee: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

Appl. No.: 17/950767

Filed: September 22, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230175709 A1	Jun. 08, 2023

Foreign Application Priority Data

KR	10-2021-0173503	Dec. 07, 2021
----	-----------------	---------------

Related U.S. Application Data

continuation parent-doc WO PCT/KR2022/012584 20220823 PENDING child-doc US 17950767

Publication Classification

Int. Cl.: **F24F1/02** (20190101); **F24F1/0011** (20190101); **F24F1/0025** (20190101); **F24F1/0047** (20190101); **F24F1/028** (20190101); **F24F1/03** (20190101); **F24F1/0317** (20190101); F24F13/14 (20060101)

U.S. Cl.:

CPC **F24F1/0025** (20130101); **F24F1/0011** (20130101); **F24F1/0047** (20190201); F24F2013/1446 (20130101); F24F2221/20 (20130101)

Field of Classification Search

CPC: F24F (1/02); F24F (1/028); F24F (1/03); F24F (1/0317); F24F (2140/40); F24F (13/222); F24F (13/22); F24F (13/10); F24F (13/08); F24F (13/082); F24F (13/084); F24F (1/0047); F24F (2210/20)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5787717	12/1997	Bang	62/157	F24F 13/20
2006/0070658	12/2005	Nakata	137/312	F24F 1/0083
2009/0241576	12/2008	Tsuji	454/358	F24F 1/0047
2010/0175418	12/2009	Choi	62/419	F24F 1/0073
2010/0323606	12/2009	Okada	454/338	F24F 1/0073

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
203628965	12/2013	CN	N/A
104266264	12/2014	CN	N/A
104896698	12/2014	CN	N/A
207501389	12/2017	CN	N/A
210179768	12/2019	CN	N/A
210486007	12/2019	CN	N/A
211316480	12/2019	CN	N/A
213872900	12/2020	CN	N/A
109405072	12/2021	CN	N/A
1 693 626	12/2016	EP	N/A
2003-262353	12/2002	JP	N/A
4039453	12/2007	JP	N/A
2012-242005	12/2011	JP	N/A
10-2003-0083076	12/2002	KR	N/A
20-0379772	12/2004	KR	N/A

OTHER PUBLICATIONS

International Search Report dated Dec. 16, 2022 in International Patent Application No.

PCT/KR2022/012584 (3 pages; 3 pages English translation). cited by applicant

PCT/ISA/237 dated Dec. 16, 2022 in International Patent Application No. PCT/KR2022/012584 (5 pages). cited by applicant

Primary Examiner: Jules; Frantz F

Assistant Examiner: Thompson; Jason N

Attorney, Agent or Firm: STAAS & HALSEY LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application, under 35 U.S.C. § 111(a), of International Application No. PCT/KR2022/012584, filed on Aug. 23, 2022, which claims priority to Korean Patent Application No. 10-2021-0173503, filed on Dec. 7, 2021 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(1) The disclosed invention relates to an air conditioner, and to an air conditioner having an improved structure.

2. Description of Related Art

(2) In general, an air conditioner is an apparatus which controls temperature, humidity, and air current distribution to levels proper for human activities and removes dust from air by using a refrigeration cycle.

(3) The air conditioner includes an outdoor unit and an indoor unit. The outdoor unit may include a compressor, an outdoor heat exchanger, an expansion device, and the like, and the indoor unit may include an indoor heat exchanger, a blower fan, and the like. An expansion device may be provided in the indoor unit.

(4) The air conditioner may be provided adjacent to a window covering, such as a curtain, a blind, or the like. However, when the window covering blocks a suction port of the air conditioner, the air conditioning performance of the air conditioner may be greatly reduced. That is, cooling or heating efficiency may decrease.

SUMMARY

(5) According to an aspect of the disclosure, there is provided an air conditioner including: a housing including a suction port to communicate with a space in which a window covering is installed and to suck air from the space between the window covering and the housing into the housing, and a discharge port through which the suctioned air is discharged to an indoor area; a heat exchanger to exchange heat with the suctioned air through the suction port; a fan to allow the air to be suctioned through the suction port or to allow the suctioned air to be discharged through the discharge port; and a guide panel at the suction port to guide the window covering to prevent the window covering from covering the suction port.

(6) The guide panel may be movable between a first position and a second position. In the second position, the guide panel may protrude from the first position toward the window covering to push the window covering away from the suction port.

- (7) The guide panel may be in the first position while the air conditioner is not in operation and the guide panel may be in the second position while the air conditioner is in operation.
- (8) The guide panel may be rotatable between the first position and the second position.
- (9) The guide panel may be movable forward and backward between the first position and the second position.
- (10) The housing may be installable between a ceiling inner wall and an indoor-side ceiling spaced downward from the ceiling inner wall, the air conditioner further including a discharge panel installable on the indoor-side ceiling to correspond to the discharge port, the discharge panel including a blade to open and close the discharge port.
- (11) The discharge panel may be exposed downward of the indoor-side ceiling, and the housing may be covered by the indoor-side ceiling.
- (12) The air conditioner may further include a drain cover at a lower side of the heat exchanger to collect condensed water generated in the heat exchanger, the drain cover covered by the indoor-side ceiling.
- (13) The guide panel, in the first position, may be arranged in a vertical direction, and in the second position, may be arranged in a horizontal direction.
- (14) The air conditioner may further include a motor configured to transmit a rotational force to the guide panel, wherein the guide panel may further include a motor coupling part rotatably couplable to the motor.
- (15) The guide panel may further include a panel body formed by a pair of long sides formed to extend in the first direction and a pair of short sides formed to extend in the second direction perpendicular to the first direction, and the motor coupling part on at least one of opposite side ends of the panel body along the first direction.
- (16) The housing may further include a support member configured to support the guide panel, wherein the guide panel may further include a support coupling part rotatably couplable to the support member.
- (17) Each of the support member and the support coupling part may be provided in plural, the plurality of support members are spaced apart from each other in the first direction, and the plurality of support coupling parts may be provided to correspond to the plurality of support members, respectively.
- (18) The guide panel may be arranged in a horizontal direction. In the first position, the guide panel may be accommodated in the housing. In the second position, the guide panel may come into contact with the window covering by horizontally being moved to an outside of the housing from the first position.
- (19) The air conditioner may further include: a motor configured to generate power; and a power transmission member configured to transmit the power generated by the motor to the guide panel, wherein the power transmission member may include: a pinion formed to share a rotation shaft with the motor, the pinion on one of the guide panel and the motor; and a rack gear engaged with the pinion, the rack gear provided on an other one of the guide panel and the motor.
- (20) According to another aspect of the disclosure, there is provided an air conditioner including: a housing disposed between a ceiling inner wall and an indoor-side ceiling, a suction port formed on a lateral side of the housing and communicating with a curtain box, a heat exchanger provided inside the housing and configured to exchange heat with air suctioned through the suction port, a discharge port formed at a lower side of the housing and configured to discharge the air having heat exchanged with the heat exchanger, a discharge panel configured to guide the air discharged from the discharge port and exposed downward of the indoor-side ceiling, and a guide panel provided at the suction port, the guide panel configured to guide a curtain installed in the curtain box for the curtain to be prevented from interfering with a flow of air suctioned through the suction port.
- (21) The guide panel may include a first side rotatably coupled to the housing and a second side opposite to the first side, and the guide panel may be provided to be rotatable between a first

position in which the second side is arranged adjacent to the housing and a second position in which the second side is arranged outside and pushes the curtain in a direction away from the suction port.

(22) The guide panel may include a first side facing an inside of the housing and a second side opposite to the first side, and the guide panel may be provided to be rotatable between a first position in which the second side is accommodated in the housing and a second position in which the second side is arranged outside and pushes the curtain in a direction away from the suction port.

(23) The air conditioner may further include a motor configured to generate a rotational force, a pinion connected to the motor and rotated by a rotational force of the motor and a rack gear disposed on the guide panel to be engaged with the pinion, the rack gear provided to be linearly moved by a rotary motion of the pinion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a perspective view illustrating an example of an air conditioner according to an embodiment.

(3) FIG. 2 is a perspective view illustrating the air conditioner of FIG. 1, seen from a different direction.

(4) FIG. 3 is a schematic cross-sectional view illustrating an example of an air conditioner according to an embodiment.

(5) FIG. 4 is a cross-sectional view schematically illustrating a state in which a guide panel of an example of an air conditioner according to an embodiment is provided in a first position.

(6) FIG. 5 is a cross-sectional view schematically illustrating a state in which a guide panel of an example of an air conditioner according to an embodiment is provided in a second position.

(7) FIG. 6 is a view illustrating a part of the air conditioner shown in FIG. 4.

(8) FIG. 7 is a view illustrating a part of the air conditioner shown in FIG. 5.

(9) FIG. 8 is a view illustrating a part of the air conditioner shown in FIG. 6, seen from a different direction.

(10) FIG. 9 is an enlarged exploded perspective view illustrating a part of the air conditioner shown in FIG. 8.

(11) FIG. 10 is an exploded perspective view illustrating a part of the air conditioner shown in FIG. 9, seen from a different direction;

(12) FIG. 11 is a cut-away view illustrating a part of the air conditioner shown in FIG. 10.

(13) FIG. 12 is a cross-sectional view schematically illustrating a state in which a guide panel of an example of the air conditioner according to an embodiment is provided in a first position.

(14) FIG. 13 is a cross-sectional view schematically illustrating a state in which a guide panel of an example of an air conditioner according to an embodiment is provided in a second position.

(15) FIG. 14 is a perspective view illustrating an example of a guide device of an air conditioner according to an embodiment.

(16) FIG. 15 is a bottom perspective view illustrating a guide device of the air conditioner shown in FIG. 12.

(17) FIG. 16 is an enlarged view illustrating the guide device of the air conditioner shown in FIG. 12.

(18) FIG. 17 is an exploded perspective view illustrating the guide device of the air conditioner shown in FIG. 16.

(19) FIG. 18 is a cut-away view of the guide device of the air conditioner shown in FIG. 17.

DETAILED DESCRIPTION

(20) Embodiments described in the specification and configurations shown in the accompanying drawings are merely examples of the disclosure, and various modifications may replace the embodiments and the drawings of the disclosure.

(21) Further, identical symbols or numbers in the drawings of the disclosure denote components or elements configured to perform substantially identical functions.

(22) Further, terms used herein are only for the purpose of describing particular embodiments and are not intended to limit to the disclosure. The singular form is intended to include the plural form as well, unless the context clearly indicates otherwise. It need to be further understood that the terms “include,” “including,” “have,” and/or “having” specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(23) When a part is referred to as being “connected to” another part, it may not only be directly connected to the other part but may in addition be indirectly connected to the other part. Similarly, when a part is referred to as being “coupled to” another part, it may not only be directly coupled to the other part but may in addition be indirectly coupled to the other part.

(24) In the description of an embodiment, it will be understood that, when a layer is referred to as being “on/under” another layer or substrate, it may be directly on/under the other layer or substrate, or one or more intervening layers may in addition be present.

(25) Further, it need to be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, the elements are not limited by the terms, and the terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and similarly, a second element could be termed a first element without departing from the scope of the disclosure. The term “and/or” includes combinations of one or all of a plurality of associated listed items.

(26) On the other hand, the terms “horizontal direction”, “vertical direction”, “front side”, “rear side”, “lateral side”, “lower side”, etc. used in the following description are defined based on the drawings, and the shape and location of each component are not limited thereto.

(27) One aspect of the disclosure provides an air conditioner with an improved aesthetic quality.

(28) Another aspect of the disclosure provides an air conditioner capable of preventing the air conditioning performance from being lowered.

(29) Another aspect of the disclosure provides an air conditioner capable of guiding a window covering for the window covering not to block a suction port. For example, referring to FIGS. 1 to 3, an X direction may be referred to as a longitudinal direction, a Y direction may be referred to as a traverse direction, and a Z direction may be referred to as a height direction. However, this is only designated with reference to the drawings for the sake of convenience of description, and is not limited thereto.

(30) A refrigeration cycle constituting an air conditioner may include a compressor, a condenser, an expansion valve, and an evaporator. The refrigeration cycle may enable circulation in a series of processes including compression-condensation-expansion-evaporation, and supply air having heat exchanged with a refrigerant. Accordingly, the air conditioner may adjust the temperature of the indoor space.

(31) An outdoor unit of the air conditioner may include a compressor and an outdoor heat exchanger. An indoor unit of the air conditioner may include an indoor heat exchanger. An expansion valve may be located in one of the indoor unit and the outdoor unit. The indoor heat exchanger and the outdoor heat exchanger serve as a condenser or an evaporator. When the indoor heat exchanger is used as a condenser, the air conditioner performs an indoor heating mode. When the indoor heat exchanger is used as an evaporator, the air conditioner performs an indoor cooling

mode.

(32) Hereinafter, an embodiment according to the disclosure will be described in detail with reference to the accompanying drawings.

(33) An air conditioner **1** according to an embodiment of the disclosure may be provided using an indoor unit of a ceiling type air conditioner. Hereinafter, the air conditioner **1** will be described.

(34) FIG. **1** is a perspective view illustrating an example of an air conditioner according to an embodiment. FIG. **2** is a perspective view illustrating the air conditioner of FIG. **1**, seen from a different direction. FIG. **3** is a schematic cross-sectional view illustrating an example of an air conditioner according to an embodiment.

(35) Referring to FIGS. **1** to **3**, the air conditioner **1** may include a main body **10**. The main body **10** may form the external appearance of the air conditioner **1**.

(36) The main body **10** may include a housing **10a** and a discharge panel **13** detachably coupled to a lower portion of the housing **10a**.

(37) The housing **10a** may be disposed between a ceiling inner wall **2** and an indoor-side ceiling **3** spaced downward from the ceiling inner wall **2**. The air conditioner **1** may be disposed in a first space S1 formed by the ceiling inner wall **2** and the indoor-side ceiling **3**.

(38) The housing **10a** may accommodate components of the air conditioner **1**. For example, the housing **10a** may accommodate a heat exchanger **20** and a blower fan (also referred to as a fan) **30** to be described below.

(39) For example, the housing **10a** may include a first housing **11** and a second housing **12**. The second housing **12** may be provided at a lower side of the first housing **11**. The second housing **12** may be detachably coupled to a lower portion of the first housing **11**. The first housing **11** and the second housing **12** may be provided as separate components and assembled with each other. However, the disclosure is not limited thereto, and the first housing **11** and the second housing **12** may be provided as one body.

(40) The main body **10** may include a suction port **14** and a discharge port **15**. For example, the housing **10a** may include the suction port **14** and the discharge port **15**.

(41) The suction port **14** may suction indoor air into the main body **10**. The suction port **14** may communicate with a second space S2 recessed from the indoor-side ceiling **3** toward the ceiling inner wall **2**. The suction port **14** may be provided to suction air from the second space S2.

(42) The discharge port **15** may discharge heat-exchanged air into an indoor room. The discharge port **15** may be opened and closed by a blade **16** to be described below.

(43) The suction port **14** may be provided with a grille (not shown) and/or a filter member (not shown) for filtering out dust in the air suctioned to the suction port **14**.

(44) The discharge port **15** may be provided with a wind direction control member (not shown) capable of adjusting a leftward and rightward direction (e.g., a Y direction) of the discharged air.

(45) Referring to FIG. **3**, the air conditioner **1** may suction air from the second space S2, cool or heat the suctioned air, and then discharge the air into the indoor area.

(46) For example, the air conditioner **1** may suction air in a lateral direction of the housing **10a** and discharge the air in a lower direction of the housing **10a** (see the arrow in FIG. **3**). Here, the lateral direction of the housing **10a** may approximately correspond to -X direction in FIG. **3**. The lower direction of the housing **10a** may approximately correspond to -Z direction in FIG. **3**. The air conditioner **1** may suction air from one side connecting an upper portion to a lower portion of the housing **10a**. The air conditioner **1** may discharge air from a lower portion of the housing **10a**. However, the disclosure is not limited thereto, the air conditioner **1** may be referred to as suctioning air from the rear side of the housing **10a** and discharging the air to the front lower side of the housing **10a** with respect to the flow of air.

(47) The discharge panel **13** may be provided to correspond to the discharge port **15**. The discharge panel **13** may be installed on the indoor-side ceiling **3**. The discharge panel **13** may be detachably mounted on the indoor-side ceiling **3**. The discharge panel **13** may be exposed downward of the

indoor-side ceiling **3**. At least a portion of the discharge panel **13** may protrude downward of the indoor-side ceiling **3**.

(48) The discharge panel **13** may include a blade **16** provided to open and close the discharge port **15**. The blade **16** may guide the air discharged through the discharge port **15**. The blade **16** may be rotated by a blade driver **16a**. For example, the blade driver **16a** may be coupled to at least one of the opposite side ends of the blade **16** in an extending direction (e.g., the Y direction) of the blade **16**. For example, the blade driver **16a** may include a step motor.

(49) For example, the blade **16** may have a substantially rectangular shape including a pair of long sides and a pair of short sides. The blade **16** may have an approximately plate shape.

(50) The discharge panel **13** may include a panel body **13a** provided to cover an edge of the blade **16**. For example, the panel body **13a** of the discharge panel **13** may include a front portion **13b** facing forward (e.g., -X direction) with respect to the air discharge direction, a rear portion **13c** facing rearward (e.g., +X direction) with respect to the air discharge direction, and side portions **13d** connecting the front portion **13b** to the rear portion **13c**.

(51) The discharge panel **13** may include a support rib **18** provided to protrude from the panel body **13a** toward the first space S1. The support rib **18** may rotatably support the blade **16**.

(52) The support ribs **18** may extend upwardly from the panel body **13a**. The support rib **18** may be provided in the first space S1.

(53) The support rib **18** may extend to connect the front portion **13b** to the rear portion **13c** of the panel body **13a**. The support rib **18** may reinforce the strength of the panel body **13a**. The support rib **18** may prevent the panel body **13a** from sagging.

(54) For example, the support rib **18** may be provided in plural, and the plurality of support ribs **18** may be provided to be spaced apart from each other along the extending direction of the blade **16**.

(55) A heat exchanger **20** may be provided to allow air suctioned through the suction port **14** to heat-exchange with a refrigerant. The heat exchanger **20** may be accommodated in the housing **10a**. For example, the heat exchanger **20** may include a tube in which a refrigerant flows and a heat exchange fin coming in contact with the tube to enlarge a heat transfer area. For example, the heat exchanger **20** may be slantingly disposed substantially perpendicular to the direction of the air flow.

(56) A blower fan **30** may force air to flow. The blower fan **30** may allow indoor air to be suctioned through the suction port **14** or allow heat-exchanged air to be discharged back into the indoors through the discharge port **15**. The blower fan **30** may be accommodated in the housing **10a**. The blower fan **30** may be driven by a fan motor (not shown).

(57) A drain cover **19** for collecting condensed water generated in the heat exchanger **20** may be provided at a lower side of the heat exchanger **20**. The drain cover **19** may include a drain tray part **19a** in which condensed water is accommodated. For example, the condensed water accommodated in the drain cover **19** may be drained to the outside through a drain hose (not shown) or the like. The drain cover **19** may be provided to support the heat exchanger **20**. The drain cover **19** may further include a heat insulating material **19b** provided to insulate the heat exchanged air.

(58) The drain cover **19** may be covered by the indoor-side ceiling **3**. The drain cover **19** may be disposed between the ceiling inner wall **2** and the indoor-side ceiling **3**. The drain cover **19** may be provided in the first space S1. The drain cover **19** may be prevented from being exposed downward of the indoor-side ceiling **3**.

(59) In general, in the case of a ceiling-type air conditioner, a ceiling panel communicating with each of a suction port and a discharge port of a main body may be mounted on an indoor-side ceiling. In this case, the ceiling panel may include a suction panel corresponding to the suction port and a discharge panel corresponding to the discharge port. The ceiling panel suctions air from a lower side thereof, cools or heats the air, and then discharges the cooled or heated air to the lower side again. Such a ceiling panel is installed on the indoor-side ceiling while protruding downward from a ceiling surface of the indoor-side ceiling. In addition, the ceiling panel needs to secure an

area for communicating with each of the suction port and discharge port of the main body, and has a large size. In other words, as the area occupied by the ceiling panel in the indoor-side ceiling increases, the area of a portion protruding from the ceiling surface of the ceiling panel may also increase. As a result, the ceiling panel may spoil the aesthetics of indoors.

(60) On the other hand, in the air conditioner **1** according to an embodiment, the indoor aesthetics may be improved by reducing the area of the ceiling panel. The housing **10a** of the air conditioner **1** may be disposed in the first space S1 formed between the ceiling inner wall **2** and the indoor-side ceiling **3**. The housing **10a** may suction air in a lateral direction from the second space S2 recessed from the indoor-side ceiling **3** toward the ceiling inner wall **2**. The housing **10a** may allow the air suctioned from the second space S2 and heat-exchanged to be discharged in the downward direction. The discharge panel **13** may be coupled to a lower portion of the housing **10a** to guide the discharge air, and may be mounted on the indoor-side ceiling **3**. That is, the housing **10a** may be covered by the indoor-side ceiling **3** and the discharge panel **13**, with only the discharge panel **13** seen by the user when the user views the indoor-side ceiling **3** from the indoors. In other words, only the discharge panel **13** corresponding to the discharge port **15** may be provided on the indoor-side ceiling **3**. That is, since the air conditioner **1** may reduce an area by as much as a portion (e.g., a suction panel) corresponding to the suction port in the existing ceiling panel, the proportion of the ceiling panel protruding from a ceiling surface **3a** may also be reduced. In addition, in the air conditioner **1**, only the discharge panel **13** having a relatively neat appearance is exposed, improving aesthetics and appearance quality. In addition, in terms of interior design, excellent aesthetics may be formed.

(61) Meanwhile, the second space S2 may be a space recessed or opened from the indoor-side ceiling **3** toward the ceiling inner wall **2**.

(62) For example, referring to FIG. **3**, a window covering **6** (e.g., a shading member, such as a curtain or blind) may be installed in the second space S2. The window covering **6** may be installed in the second space S2 by an installation member **6a**. For example, the installation member **6a** may include a rail and/or a bar, but is not limited thereto, and may be provided as a generally known type of member for installing the window covering **6**. The second space S2 may include a curtain box. The second space S2 may be adjacent to a window frame **7** and a window **8**, but is not limited thereto, and may be provided to be adjacent to a wall of an indoor space.

(63) The first space S1 and the second space S2 may communicate with each other. For example, an opening **5** for communicating the first space S1 and the second space S2 with each other may be provided. The opening **5** may be formed by cutting at least a portion of a vertical panel **4** extending upward from the indoor-side ceiling **3**.

(64) Meanwhile, the indoor-side ceiling **3** may be referred to as a first wall forming the ceiling surface **3a**, and the vertical panel **4** may be referred to as a second wall extending upwardly from the first wall. For example, the indoor-side ceiling **3** and the vertical panel **4** may be integrally formed with each other. However, this is merely illustrative, and the indoor-side ceiling **3** and the vertical panel **4** may be provided as separate components.

(65) The second space S2 may be formed by the ceiling inner wall **2** and the vertical panel **4**. However, the disclosure is not limited thereto. For example, a horizontal panel (not shown) extending in the horizontal direction from the vertical panel **4** may be additionally provided. The horizontal panel may be provided at a position higher than that of the indoor-side ceiling **3**. That is, the horizontal panel may be spaced downwardly from the inner ceiling wall **2** by an interval smaller than the interval between the ceiling inner wall **2** and the indoor-side ceiling **3**. In this case, the second space S2 may be formed by the horizontal panel and the vertical panel **4**. The horizontal panel may be referred to as a third wall extending laterally from the second wall.

(66) Referring to FIG. **3**, the air conditioner **1** may include a guide device **40**. The guide device **40** may guide the window covering **6**. The guide device **40** may prevent the window covering **6** from covering the suction port **14**. The guide device **40** may be referred to as a guide assembly.

(67) The guide device **40** may be provided at the suction port **14**. The guide device **40** may be detachably mounted on the housing **10a** to be adjacent to the suction port **14**.

(68) The air conditioner may be disposed adjacent to a space in which window covering is installed. In this case, during driving (operation) of the air conditioner, the window covering may be suctioned into the suction port by the suction force generated from the suction port. Accordingly, the window covering may block the suction port, so that the suction power at the suction port is reduced, and indoor air may not be smoothly introduced into the inside of the air conditioner. Accordingly, the air conditioning performance of the air conditioner may be lowered. That is, the cooling and heating efficiency of the air conditioner may be lowered.

(69) In contrast, the air conditioner **1** according to the embodiment may include the guide device **40**, to prevent the window covering **6** from blocking the suction port **14**. The guide device **40** may guide the window covering **6** to prevent the window covering **6** from moving toward the suction port **14**. Accordingly, the indoor air is not obstructed by the window covering **6** when flowing into the suction port **14**, and thus may be smoothly introduced into the air conditioner. That is, degradation of the air conditioning performance of the air conditioner **1** due to the window covering **6** may be prevented. Degradation in the cooling/heating efficiency of the air conditioner **1** due to the window covering **6** may be prevented.

(70) FIG. **4** is a cross-sectional view schematically illustrating a state in which a guide panel of an example of an air conditioner according to an embodiment is provided in a first position. FIG. **5** is a cross-sectional view schematically illustrating a state in which a guide panel of an example of an air conditioner according to an embodiment is provided in a second position. FIG. **6** is a view illustrating a part of the air conditioner shown in FIG. **4**. FIG. **7** is a view illustrating a part of the air conditioner shown in FIG. **5**. FIG. **8** is a view illustrating the part of the air conditioner shown in FIG. **6** seen from a different direction. FIG. **9** is an enlarged exploded perspective view illustrating the part of the air conditioner shown in FIG. **8**. FIG. **10** is an exploded perspective view illustrating the part of the air conditioner shown in FIG. **9** seen from a different direction. FIG. **11** is a cut-away view illustrating the part of the air conditioner shown in FIG. **10**.

(71) Hereinafter, an example **40a** of the guide device **40** will be described with reference to FIGS. **4** to **11**. The guide device **40a** may be rotatably provided.

(72) The guide device **40a** may include a guide panel **110**.

(73) The guide panel **110** may guide the window covering **6** to prevent the window covering **6** from covering the suction port **14**. The guide panel **110** may guide the window covering **6** to prevent the window covering **6** from interfering with the flow of air suctioned through the suction port **14**. The guide panel **110** may be provided to secure a space between the suction port **14** and the window covering **6**.

(74) For example, the guide panel **110** may prevent the window covering **6** from moving toward the suction port **14**. The guide panel **110** may push the window covering **6** in a direction away from the suction port **14**. The guide panel **110** may move the window covering **6** away from the main body **10** of the air conditioner **1** by applying a predetermined force to the window covering **6**. The guide panel **110** may be provided to push the window covering **6** toward the window **8**.

(75) The guide panel **110** may be provided at the suction port **14**. The guide panel **110** may be detachably mounted on the housing **10a** to be disposed adjacent to the suction port **14**.

(76) The guide panel **110** may include a panel body **111**. The panel body **111** may form the external appearance of the guide panel **110**.

(77) The panel body **111** may include a pair of long sides **114** and a pair of short sides **115**. The panel body **111** may have a substantially rectangular shape. The panel body **111** may have a substantially plate shape.

(78) The panel body **111** may include a first side **116** rotatably coupled to the housing **10a** and a second side **117** opposite to the first side **116**. For example, the first side **116** may be a long side **114** adjacent to the heat exchanger **20** among the pair of long sides **114**, and the second side **117**

may be the other long side **114** among the pair of long sides **114**. That is, the first side **116** may be closer to the heat exchanger **20** than the second side **117**.

(79) The guide panel **110** may include a motor coupling part **112**. The motor coupling part **112** may be coupled to a motor **120**. The motor coupling part **112** may be directly coupled to the motor **120** or may be indirectly coupled to the motor **120** by another element (e.g., a power transmission member **130** to be described below). The motor coupling part **112** may receive rotational force from the motor **120**.

(80) The motor coupling part **112** may extend from one side of the panel body **111**. For example, the motor coupling part **112** may be disposed adjacent to the first side **116**. The motor coupling part **112** may be formed on the first side **116**. The motor coupling part **112** may be disposed adjacent to the short side **115**. The motor coupling part **112** may be formed on the short side **115**. The motor coupling part **112** may be formed at a corner in which the first side **116** meets the short side **115**. However, the motor coupling part **112** is not limited to the above-described examples as long as it is provided on a portion of the panel body **111** to be coupled to the motor, and the arrangement and shape thereof are not limited to the examples shown in the drawings.

(81) The motor coupling part **112** may be provided on at least one of opposite side ends of the panel body **111** in a direction along the long side of the panel body **111**.

(82) For example, the motor coupling part **112** may include an insertion groove **118**. The insertion groove **118** may be provided to accommodate at least a portion of an elastic member **150** to be described below. The elastic member **150** may be inserted into the insertion groove **118**.

(83) The guide panel **110** may include a support coupling part **113**. The support coupling part **113** may be rotatably coupled to the housing **10a**. The support coupling part **113** may be rotatably coupled to a support member **17** formed in the housing **10a**. The support member **17** may rotatably support the support coupling part **113**.

(84) Referring to FIG. **9**, the support coupling part **113** may include a support protrusion **113a** corresponding to a support hole **17a** formed in the support member **17**. The support protrusion **113a** may be rotatably coupled to the support hole **17a**. However, the disclosure is not limited thereto, and contrary to the above-described example, the support coupling part **113** may include a support hole, and the support member **17** may include a support protrusion.

(85) The support coupling part **113** may extend from one side of the panel body **111**. For example, the support coupling part **113** may be disposed adjacent to the first side **116**. The support coupling part **113** may be formed on the first side **116**.

(86) The support coupling part **113** may be provided in plural. For example, the plurality of support coupling parts **113** may be disposed to be spaced apart from each other along the long side direction of the panel body **111**. The support member **17** of the housing **10a** may also be provided in plural to correspond to the support coupling parts **113**.

(87) The guide panel **110** may be rotatably provided. The guide panel **110** may be rotated by receiving rotational force generated from the motor **120**.

(88) Referring to FIGS. **4** to **7**, the guide panel **110** may be provided to be movable between a first position P1 and a second position P2. The guide panel **110** may be provided to be rotatable between the first position P1 and the second position P2.

(89) When the guide panel **110** is in the first position P1, the guide panel **110** may be provided not to be interfered with the window covering **6**. For example, when the air conditioner **1** is not in operation, the guide panel **110** may be provided in the first position P1.

(90) When the air conditioner **1** is not in operation, the suction port **14** does not suction air, so the window covering **6** may not be suctioned into the suction port **14**. Accordingly, the guide panel **110** may not need to push the window covering **6** in a direction away from the suction port **14**. In other words, when the guide device **40** is not used (that is, when the air conditioner **1** does not operate), the guide panel **110** may be provided in the first position P1 without being protruded or exposed to the second space S2 in consideration of the interior aesthetics.

(91) In addition, when the air conditioner **1** is not in operation, the guide panel **110** may be provided in the first position P1 so as not to interfere with the operation of the window covering **6** installed in the second space S2. For example, when the window covering **6** is a curtain, the guide panel **110**, in response to non-operation of the air conditioner **1**, may be provided in the first position P1 so as not to interfere with the folding/unfolding operation of the curtain **6**.

(92) For example, referring to FIGS. **4** and **6**, the guide panel **110**, when provided in the first position P1, may be arranged in a vertical direction (the Z direction). When the guide panel **110** is in the first position P1, the short side **115** of the panel body **111** may be arranged in the vertical direction. When the guide panel **110** is in the first position P1, the guide panel **110** may be provided to be in close contact with the vertical panel **4**. When the guide panel **110** is in the first position P1, the second side **117** of the panel body **111** may be arranged adjacent to the housing **10a** than when the guide panel **110** is in the second position P2. In the drawings, the guide panel **110**, when provided in the first position P1, is illustrated as being disposed outside the housing **10a**, but the disclosure is not limited thereto, and the guide panel **110**, when provided in the first position P1, may be accommodated inside the housing **10a**.

(93) When the guide panel **110** is in the second position P2, the guide panel **110** may be provided to interfere with the window covering **6**. For example, when the air conditioner **1** operates, the guide panel **110** may be provided in the second position P2.

(94) As described above, when the air conditioner **1** operates, the window covering **6** may be moved toward the suction port **14** by the suction force generated from the suction port **14**. In this case, the window covering **6** may cover the suction port **14**, causing a portion of a suction passage to be blocked. Accordingly, the flow rate of the air suctioned into the air conditioner **1** may decrease, and the air conditioning performance may be degraded.

(95) During operation of the air conditioner **1**, the guide panel **110** may be provided in the second position P2 and prevent the window covering **6** from covering the suction port **14**. Since the guide panel **110** is provided in the second position P2, the window covering **6** may be prevented from being suctioned into the suction port **14**. Since the guide panel **110** is provided in the second position P2, a certain amount of space may be secured between the suction port **14** and the window covering **6**. That is, the guide panel **110** may be provided in the second position P2 to secure a flow path through which air flows to the suction port **14**. Accordingly, degradation of the air conditioning performance of the air conditioner **1** may be prevented.

(96) When the guide panel **110** is in the second position P2, the guide panel **110** may be in a state of being protruded toward the window covering **6** from the first position P1. When the guide panel **110** is in the second position P2, the guide panel **110** may push the window covering **6** in a direction away from the suction port **14**. When the guide panel **110** is in the second position P2, the guide panel **110** may apply a force to the window covering **6** in a direction away from the suction port **14**. When the guide panel **110** is in the second position P2, the guide panel **110** may push the window covering **6** toward the window **8**.

(97) For example, referring to FIGS. **5** and **7**, when the guide panel **110** is in the second position P2, the guide panel **110** may be arranged in a horizontal direction (the X direction). When the guide panel **110** is in the second position P2, the short side **115** of the panel body **111** may be arranged in a horizontal direction. When the guide panel **110** is in the second position P2, the second side **117** of the panel body **111** may be disposed to be farther from the housing **10a** than when the guide panel **110** is in the first position P1. In this case, the second side **117** may be disposed outside the housing **10a** while coming in contact with the window covering **6**.

(98) The guide device **40a** may include the motor **120** for driving the guide panel **110**. The motor **120** may generate power. The motor **120** may generate a rotational force. The motor **120** may transmit a rotational force to the guide panel **110**. For example, the motor **120** may be a step motor. However, the disclosure is not limited thereto, and the motor **120** may be provided with various driving sources capable of driving the guide panel **110**.

(99) The motor **120** may include a motor shaft **121**. The motor shaft **121** may form a center of rotation.

(100) For example, the motor shaft **121** may include a cut portion **122**. The motor shaft **121** may have a cross section in a shape of a circle that is partially cut. Accordingly, the motor shaft **121** may transmit the power generated from the motor **120** to the guide panel **110** without slip by including the cut portion **122**. However, the shape of the motor shaft **121** is not limited thereto, and the cross-section thereof may be provided in a shape of a circle without the cut portion **122**.

(101) The guide device **40a** may include the power transmission member **130**. The power transmission member **130** may transmit the rotational force generated by the motor **120** to the guide panel **110**. However, the disclosure is not limited thereto, and the guide device **40a** may not be separately provided with the power transmission member **130**, and the rotational force generated by the motor **120** may be directly transmitted to the guide panel **110**. In this case, the motor **120** and the guide panel **110** may be directly connected to each other.

(102) For example, the power transmission member **130** may include a connecting member **140**. For example, the power transmission member **130** may include an elastic member **150**.

(103) The connecting member **140** may connect the motor **120** and the guide panel **110** to each other. When the guide device **40a** includes the elastic member **150**, the connecting member **140** may connect the motor **120** and the elastic member **150** to each other.

(104) The connecting member **140** may transmit power received from the motor **120** to the guide panel **110**. When the guide device **40a** includes the elastic member **150**, the connecting member **140** may transmit power received from the motor **120** to the elastic member **150**.

(105) The connecting member **140** may be provided between the motor **120** and the guide panel **110** to prevent mutual interference between the motor **120** and the guide panel **110**. For example, when the motor **120** is directly connected to the guide panel **110**, the guide panel **110** may directly receive a rotational force from the motor **120** and therefore may be damaged by vibration, impact, etc. generated during rotation of the motor **120**. The connecting member **140** prevents the guide panel **110** and the motor **120** from being directly connected to each other, thereby improving durability, abrasion resistance, and the like of the guide device **40a**. That is, the connecting member **140** may promote smooth rotation of the guide panel **110**.

(106) For example, the connecting member **140** may include a polyoxymethylene (POM) material. The connecting member **140** may be lubricated.

(107) The connecting member **140** may include a connecting body **141**. The connecting body **141** may form the external appearance of the connecting member **140**.

(108) The connecting member **140** may include a motor connecting part **143** provided to be connected to the motor **120**. The motor connecting part **143** may include a shaft insertion groove **145** into which the motor shaft **121** is inserted. The shaft insertion groove **145** may have a shape corresponding to the cut portion **122** of the motor shaft **121**.

(109) The connecting member **140** may include an extension part **142** provided on the opposite side of the motor connecting part **143** and extending along the motor shaft **121**. The extension part **142** may extend in a direction along the long side **114** of the panel body **111** of the guide panel **110**.

(110) The extension part **142** may be connected to the guide panel **110**. The extension part **142** may be coupled to the elastic member **150** to thereby be connected to the guide panel **110**. For example, the extension part **142** may have a polygonal prism shape. For example, the extension part **142** may include an approximately “X” shape. However, the disclosure is not limited thereto, and may include various shapes for smoothly transmitting power.

(111) The extension part **142** may include a coupling groove **144** to which a coupling protrusion **152** of the elastic member **150** to be described below is coupled. The coupling groove **144** may be provided in a shape corresponding to that of the coupling protrusion **152**. As the coupling protrusion **152** is inserted into the coupling groove **144**, the elastic member **150** may be fixed to the extension part **142**. When the guide device **40a** rotates, the extension part **142** and the elastic

member **150** may not be easily separated from each other. For example, the coupling groove **144** may be formed on the outer surface of the extension part **142**.

(112) The elastic member **150** may be provided between the guide panel **110** and the motor **120**. The elastic member **150** may be provided between the guide panel **110** and the connecting member **140**. The elastic member **150** may be provided to reduce noise and vibration when the guide panel **110** is rotated. The elastic member **150** may include an elastic material. For example, the elastic member **150** may include at least one of rubber or silicone. However, this is only an example, and the elastic member **150** may be provided in various forms as long as it includes a material having elasticity.

(113) The elastic member **150** may be coupled to the guide panel **110**. The elastic member **150** may be inserted into the insertion groove **118** of the motor coupling part **112**. For example, the elastic member **150** may be inserted into the insertion groove **118** in an interference fit manner.

(114) The elastic member **150** may transmit power received from the motor **120** to the guide panel **110**. The elastic member **150** may transmit power received from the connecting member **140** to the guide panel **110**.

(115) The elastic member **150** may include a hollow part **151**. The extension part **142** of the connecting member **140** may be accommodated in the hollow part **151**. The extension part **142** of the connecting member **140** may be inserted into the hollow part **151**. The hollow part **151** may include a shape corresponding to that of the extension part **142**. The hollow part **151** may include an approximately “X” shape.

(116) The hollow part **151** of the elastic member **150** may be provided with the coupling protrusion **152**. The coupling protrusion **152** may protrude from the inner surface of the elastic member **150**. The coupling protrusion **152** may be coupled to the coupling groove **144** of the connecting member **140**. The coupling protrusion **152** may be inserted into the coupling groove **144**. As the coupling protrusion **152** is inserted into the coupling groove **144**, the elastic member **150** may be fixed to the extension part **142**. Accordingly, when the guide device **40a** rotates, the extension part **142** and the elastic member **150** may not be easily separated from each other.

(117) Meanwhile, in the drawings, the connecting member **140** is illustrated as including the coupling groove **144** and the elastic member **150** is illustrated as including the coupling protrusion **152**, but the disclosure is not limited thereto. For example, the connecting member **140** may include a coupling protrusion protruding from the outer surface of the connecting body **141**, and the elastic member **150** may include a coupling groove recessed from the inner surface of the hollow part **151**.

(118) The elastic member **150** may have a polygonal column shape. For example, the elastic member **150** may have a hexagonal pillar shape. However, the disclosure is not limited thereto, and the elastic member **150** may include various shapes.

(119) FIG. **12** is a cross-sectional view schematically illustrating a state in which a guide panel of an example of the air conditioner according to an embodiment is provided in a first position. FIG. **13** is a cross-sectional view schematically illustrating a state in which a guide panel of an example of an air conditioner according to an embodiment is provided in a second position. FIG. **14** is a perspective view illustrating an example of a guide device of an air conditioner according to an embodiment. FIG. **15** is a bottom perspective view illustrating a guide device of the air conditioner shown in FIG. **12**. FIG. **16** is an enlarged view illustrating the guide device of the air conditioner shown in FIG. **12**. FIG. **17** is an exploded perspective view illustrating the guide device of the air conditioner shown in FIG. **16**. FIG. **18** is a cut-away view of the guide device of the air conditioner shown in FIG. **17**.

(120) Next, an example **40b** of the guide device **40** will be described with reference to FIGS. **12** to **17**. The air conditioner **1** shown in FIGS. **12** to **17** may include the same construction as that of the air conditioner **1** described above, except that the guide device **40b** is configured to move forward and backward. The same reference numerals may be assigned to the same components, and the

same descriptions may be omitted.

(121) The guide device **40b** may include a guide panel **210**.

(122) The guide panel **210** may guide the window covering **6** to prevent the window covering **6** from covering the suction port **14**. The guide panel **210** may guide the window covering **6** to prevent the window covering **6** from interfering with the flow of air suctioned through the suction port **14**. The guide panel **210** may be provided to secure a space between the suction port **14** and the window covering **6**.

(123) For example, the guide panel **210** may prevent the window covering **6** from moving toward the suction port **14**. The guide panel **210** may push the window covering **6** in a direction away from the suction port **14**. The guide panel **210** may move the window covering **6** away from the main body **10** of the air conditioner **1** by applying a predetermined force to the window covering **6**. The guide panel **210** may be provided to push the window covering **6** toward the window **8**.

(124) The guide panel **210** may be provided at the suction port **14**. The guide panel **210** may be detachably mounted on the housing **10a** to be disposed adjacent to the suction port **14**.

(125) The guide panel **210** may include a panel body **211**. The panel body **211** may form the external appearance of the guide panel **210**.

(126) The panel body **211** may include a pair of long sides **214** and a pair of short sides **215**. The panel body **211** may have a substantially rectangular shape. The panel body **211** may have a substantially plate shape.

(127) The panel body **211** may include a first side **216** rotatably coupled to the housing **10a** and a second side **217** opposite to the first side **216**. For example, the first side **216** may be provided to face the inside of the housing **10a**, and the second side **217** may be provided to face the outside of the housing **10a**. The second side **217** may be provided to face the window covering **6**. For example, the first side **216** may be a long side **214** adjacent to the heat exchanger **20** among the pair of long sides **214**, and the second side **217** may be the other long side **214** among the pair of long sides **214**. That is, the first side **216** may be closer to the heat exchanger **20** than the second side **217**.

(128) The guide panel **210** may be arranged in a horizontal direction. For example, the guide panel **210** may be arranged to correspond to the X direction. For example, the short side **215** of the panel body **211** may be arranged in a horizontal direction to correspond to the X direction.

(129) The guide panel **210** may be provided to enable forward and backward movement. The guide panel **210** may be provided to be movable in the X direction. The guide panel **210** may be provided to be movable in a horizontal direction. The guide device **40b** may be provided to enable linear motion.

(130) The guide panel **210** may be provided to be movable forward and backward by receiving power generated from a motor **220**. The guide panel **210** may receive rotational force from the motor **220** to move forward (−X direction) or rearward (+X direction).

(131) Referring to FIGS. **12** and **13**, the guide panel **210** may be provided to be movable between a first position P1 and a second position P2. The guide panel **210** may be provided to be horizontally movable between the first position P1 and the second position P2. The guide panel **210** may move linearly between the first position p1 and the second position P2.

(132) When the guide panel **210** is in the first position P1, the guide panel **210** may be provided so as not to interfere with the window covering **6**. For example, when the air conditioner **1** is not in operation, the guide panel **210** may be provided in the first position P1.

(133) When the air conditioner **1** is not in operation, the suction port **14** does not suction air, so the window covering **6** may not be suctioned into the suction port **14**. Accordingly, the guide panel **210** may not need to push the window covering **6** in a direction away from the suction port **14**. In other words, when the guide device **40** is not used (that is, when the air conditioner **1** does not operate), the guide panel **210** may be provided in the first position P1 without being protruded or exposed to the second space S2 in consideration of the interior aesthetics.

(134) In addition, when the air conditioner **1** is not in operation, the guide panel **210** may be provided in the first position P1 so as not to interfere with the operation of the window covering **6** installed in the second space S2. For example, when the window covering **6** is a curtain, the guide panel **210**, in response to non-operation of the air conditioner **1**, may be provided in the first position P1 so as not to interfere with the folding/unfolding operation of the curtain **6**.

(135) For example, referring to FIG. **12**, the guide panel **210**, when provided in the first position P1, may be accommodated in the housing **10a**. The guide panel **210**, when provided in the first position P1, may be stored in the housing **10a**. When the guide panel **210** is in the first position P1, the second side **217** of the panel body **211** may be arranged adjacent to the housing **10a** than when the guide panel **210** is in the second position P2. When the guide panel **210** is in the first position P1, the second side **217** of the panel body **211** may be disposed inside the housing **10a**.

(136) When the guide panel **210** is in the second position P2, the guide panel **210** may be provided to interfere with the window covering **6**. For example, when the air conditioner **1** operates, the guide panel **210** may be provided in the second position P2.

(137) As described above, when the air conditioner **1** operates, the window covering **6** may be moved toward the suction port **14** by the suction force generated from the suction port **14**. In this case, the window covering **6** may cover the suction port **14**, causing a portion of a suction passage to be blocked. Accordingly, the flow rate of the air suctioned into the air conditioner **1** may decrease, and the air conditioning performance may be degraded.

(138) During operation of the air conditioner **1**, the guide panel **210** may be provided in the second position P2 and prevent the window covering **6** from covering the suction port **14**. Since the guide panel **210** is provided in the second position P2, the window covering **6** may be prevented from being suctioned into the suction port **14**. Since the guide panel **210** is provided in the second position P2, a certain amount of space may be secured between the suction port **14** and the window covering **6**. That is, the guide panel **210** may be provided in the second position P2 to secure a flow path through which air flows to the suction port **14**. Accordingly, degradation of the air conditioning performance of the air conditioner **1** may be prevented.

(139) When the guide panel **210** is in the second position P2, the guide panel **210** may be in a state of being protruded toward the window covering **6** from the first position P1. When the guide panel **210** is in the second position P2, the guide panel **210** may push the window covering **6** in a direction away from the suction port **14**. When the guide panel **210** is in the second position P2, the guide panel **210** may apply a force to the window covering **6** in a direction away from the suction port **14**. When the guide panel **210** is in the second position P2, the guide panel **210** may push the window covering **6** toward the window **8**.

(140) When the guide panel **210** is in the second position P2, the second side **217** of the panel body **211** may be arranged further away from the housing **10a** than when the guide panel **210** is in the first position P1. In this case, the second side **217** may be disposed outside the housing **10a** and come in contact with the window covering **6**.

(141) When the guide panel **210** is in the second position P2, the guide panel **210** may come in contact the window covering **6** due to horizontally moving from the first position P1 toward the outside of the housing **10a**.

(142) The guide device **40b** may include the motor **220** for driving the guide panel **210**. The motor **220** may generate power. The motor **220** may generate a rotational force. The motor **220** may transmit a rotational force to the guide panel **210**. For example, the motor **220** may be a step motor. However, the disclosure is not limited thereto, and the motor **220** may be provided with various driving sources capable of driving the guide panel **210**.

(143) The motor **220** may include a motor shaft **221**. The motor shaft **221** may form a center of rotation.

(144) For example, the motor shaft **221** may include a cut portion **222**. The motor shaft **221** may have a cross section in a shape of a circle that is partially cut. Accordingly, the motor shaft **221** may

transmit the power generated from the motor **220** to the guide panel **210** without slip by including the cut portion **222**. However, the shape of the motor shaft **221** is not limited thereto, and a cross-section thereof may be provided in a shape of a circle without the cut portion **222**.

(145) The motor **220** may be substantially the same as the motor **120** described above.

(146) The guide device **40b** may include a power transmission member **230**. The power transmission member **230** may transmit the rotational force generated by the motor **220** to the guide panel **210**. However, the disclosure is not limited thereto, and the guide device **40b** may not be separately provided with the power transmission member **230**, and the rotational force generated by the motor **220** may be directly transmitted to the guide panel **210**. In this case, the motor **220** and the guide panel **210** may be directly connected to each other.

(147) The power transmission member **230** may include a pinion **260** and a rack gear **270**.

(148) The pinion **260** and the rack gear **270** transmit the power of the motor **220** to the guide panel **210**, thereby enabling the guide panel **210** to move forward and backward. The pinion **260** and the rack gear **270** may convert a rotational motion into a linear motion.

(149) The pinion **260** may be provided on one of the guide panel **210** and the motor **220**, and the rack gear **270** may be provided on the other one of the guide panel **210** and the motor **220**.

Hereinafter, a case in which the pinion **260** is provided on the motor **220** and the rack gear **270** is provided on the guide panel **210** will be described as an example. However, the disclosure is not limited thereto, and the pinion **260** may be provided on the guide panel **210** and the rack gear **270** may be provided on the motor **220**.

(150) For example, each of the pinion **260** and the rack gear **270** may be provided in plural. The pinion **260** and the rack gear **270** may be provided corresponding in number to each other. The plurality of pinions **260** and the plurality of rack gears **270** may be arranged to be spaced apart from each other by a predetermined distance along the long side **214** of the panel body **211**. The plurality of pinions **260** and the plurality of rack gears **270** may support the guide panel **210** while guiding the movement of the guide panel **210**.

(151) The pinion **260** may share a rotation axis with the motor shaft **221** of the motor **220**. The pinion **260** may be provided to be rotated by the motor **220**. The pinion **260** may perform rotary motion.

(152) The pinion **260** may be connected to the motor shaft **221** of the motor **220**. The pinion **260** may be directly connected to the motor shaft **221** of the motor **220**, or indirectly coupled to the motor **220** by other elements (e.g., a connecting member **240** and an elastic member **250**).

(153) The pinion **260** may transmit the power of the motor **220** to the rack gear **270**. The pinion **260** may transmit the rotational force received from the motor **220** to the rack gear **270**.

(154) The pinion **260** may include a pinion body **261**. The pinion body **261** may form the external appearance of the pinion **260**. At least a portion of the outer circumferential surface of the pinion body **261** may be engaged with the rack gear **270**.

(155) For example, the pinion body **261** may be provided to have one side connected to the motor **220** and the other side connected to a shaft **280** to be described below. The one side of the pinion body **261** may be provided to be connected to the motor **220** through the extension member **240** and the elastic member **250** to be described below.

(156) The pinion **260** may include an insertion groove **262** into which the elastic member **250** is inserted. The insertion groove **262** may include a fixing protrusion **264** corresponding to a fixing groove **252** of the elastic member **250** to be described below. When the elastic member **250** is inserted into the insertion groove **262**, the fixing groove **252** and the fixing protrusion **264** may be provided to be engaged with each other. Accordingly, the fixing force between the pinion **260** and the elastic member **250** may be improved.

(157) The pinion **260** may include a shaft coupling part **263** to which the shaft **280** is coupled. The shaft coupling part **263** may be provided with a shaft insertion groove **263a** into which one end of the shaft **280** may be inserted. The shaft insertion groove **263a** may include a shape corresponding

to that of a cut portion **281** of the shaft **280**, which will be described below.

(158) The rack gear **270** may be provided to engage the pinion **260**. The rack gear **270** may be provided on the guide panel **210**. The rack gear **270** may be disposed on the panel body **211** of the guide panel **210**. The rack gear **270** may extend in a direction along the short side **215** of the panel body **211**. For example, the rack gear **270** may be integrally formed with the panel body **211**.

(159) The rack gear **270** may receive power from the pinion **260**. The rack gear **270** may be provided to linearly move by the rotational movement of the pinion **260**. The rack gear **270** may be provided to be movable forward and backward. The guide panel **210** may be movable in association with the rack gear **270**.

(160) The power transmission member **230** may include the shaft **280**.

(161) The shaft **280** may be coupled to the shaft coupling part **263** of the pinion **260**. The shaft **280** may be provided coaxially with the motor shaft **221**. The shaft **280** may form a center of rotation of the pinion **260**. The shaft **280** may be coupled to the center of the pinion **260**.

(162) The shaft **280** may extend in a direction along the long side of the panel body **211**.

(163) For example, when the pinion **260** is provided in plural, the shaft **280** may be provided to connect the plurality of pinions **260** to each other. In this case, the shaft **280** may be provided in plural. However, the disclosure is not limited thereto, and one shaft **280** may be provided to connect the plurality of pinions **260**. For example, the shaft **280** may include the cut portion **281**. The shaft **280** may have a cross section in a shape of a circle that is partially cut. Accordingly, the shaft **280** may transmit the power received from the pinion **260**, which is connected at one end thereof, to the pinion **260**, which is connected at the other end thereof, without slip.

(164) For example, the power transmission member **230** may include the connecting member **240**. For example, the power transmission member **230** may include the elastic member **250**.

(165) The connecting member **240** may connect the motor **220** and the pinion **260** to each other. When the guide device **40b** includes the elastic member **250**, the connecting member **240** may connect the motor **220** and the elastic member **250** to each other.

(166) The connecting member **240** may transmit power received from the motor **220** to the pinion **210**. When the guide device **40b** includes the elastic member **250**, the connecting member **240** may transmit power received from the motor **220** to the elastic member **250**.

(167) The connecting member **240** may be provided between the motor **220** and the pinion **260** to prevent mutual interference between the motor **220** and the pinion **260**. For example, when the motor **220** is directly connected to the pinion **260**, the pinion **260** may directly receive a rotational force from the motor **220** and thus be damaged by vibration, impact, etc. generated during rotation of the motor **220**. The connecting member **240** prevents the pinion **260** and the motor **220** from being directly connected to each other, thereby improving durability, abrasion resistance, and the like of the guide device **40b**. That is, the connecting member **240** may promote smooth rotation of the pinion **260**.

(168) For example, the connecting member **240** may include a polyoxymethylene (POM) material. The connecting member **240** may be lubricated.

(169) The connecting member **240** may include a connecting body **241**. The connecting body **241** may form the external appearance of the connecting member **240**.

(170) The connecting member **240** may include a motor connecting part **243** provided to be connected to the motor **220**. The motor connecting part **243** may include a shaft insertion groove **245** into which the motor shaft **221** is inserted. The shaft insertion groove **245** may have a shape corresponding to the cut portion **222** of the motor shaft **221**.

(171) The connecting member **240** may include an extension part **242** provided on the opposite side of the motor connecting part **243** and extending along the motor shaft **221**. The extension part **242** may extend in a direction along the long side **214** of the panel body **211** of the guide panel **210**.

(172) The extension part **242** may be connected to the pinion **260**. The extension part **242** may be coupled to the elastic member **250** to thereby be connected to the pinion **260**. For example, the

extension part **242** may have a polygonal prism shape. For example, the extension part **242** may include an approximately “X” shape. However, the disclosure is not limited thereto, and may include various shapes for smoothly transmitting power.

(173) The extension part **242** may be coupled to a hollow part **251** of the elastic member **250**. The extension part **242** may be inserted into the hollow part **251** of the elastic member **250**.

(174) For example, the extension part **242** may have a shape in which a cross-sectional area becomes smaller toward the pinion **260**.

(175) Although not shown in the drawings, the extension part **242** may further include a coupling protrusion (not shown) so as not to be easily separated from the hollow part **251** of the elastic member **250**. Correspondingly, the hollow part of the elastic member **250** may further include a coupling groove (not shown) coupled to the coupling protrusion of the extension part **242**.

Conversely, the extension part **242** may include a coupling groove, and the elastic member **250** may include a coupling protrusion.

(176) The elastic member **250** may be provided between the pinion **260** and the motor **220**. The elastic member **250** may be provided between the pinion **260** and the connecting member **240**. The elastic member **250** may be provided to reduce noise and vibration when the pinion **260** rotates. The elastic member **250** may include an elastic material. For example, the elastic member **250** may include at least one of rubber or silicone. However, this is only an example, and the elastic member **20** may be provided in various forms as long as it includes a material having elasticity.

(177) The elastic member **250** may be coupled to the pinion **260**. The elastic member **250** may be inserted into the insertion groove **262** of the pinion **260**. For example, the elastic member **250** may be inserted into the insertion groove **262** in an interference fit manner.

(178) The elastic member **250** may transmit power received from the motor **220** to the pinion **260**. The elastic member **250** may transmit power received from the connecting member **240** to the pinion **260**.

(179) The elastic member **250** may include the hollow part **251**. The extension part **242** of the connecting member **240** may be accommodated in the hollow part **251**. The extension part **242** of the connecting member **240** may be inserted into the hollow part **251**. The hollow part **251** may have a shape corresponding to that of the extension part **242**.

(180) For example, the elastic member **250** may include the fixing groove **252**. Referring to FIGS. **17** and **18**, the fixing groove **252** may be provided to correspond to the fixing protrusion **264** of the pinion **260**. The fixing groove **252** may be formed by cutting a portion of the outer surface of the elastic member **250**. The fixing groove **252** may be provided to extend in a direction along the motor shaft **221**. When the fixing protrusion **264** of the pinion **260** and the fixing groove **252** of the elastic member **250** are engaged with each other, the fixing force between the pinion **260** and the elastic member **250** may be improved. Accordingly, the elastic member **250** may effectively reduce noise and vibration generated when the pinion **260** rotates.

(181) However, the disclosure is not limited thereto, and the elastic member **250** may include a fixing protrusion protruding from the outer surface of the elastic member **250**, and the pinion **260** may include a fixing groove corresponding to the fixing protrusion.

(182) On the other hand, although not shown in the drawings, the guide panel of the guide device **40** may be provided to be extended and contracted. When the air conditioner **1** is not in operation, the guide panel may be contracted so as not to interfere with the window covering **6**, and when the air conditioner **1** is in operation, the guide panel may be extended to interfere with the window covering **6**. For example, the guide panel may be operated in a telescopic manner.

(183) Although not shown in the drawings, the guide panel of the guide device **40** may be provided to be foldable or bendable. For example, when the air conditioner **1** is not in operation, the guide panel may be folded so as not to interfere with the window covering **6**, and when the air conditioner **1** is in operation, the guide panel may be folded to interfere with the window covering **6**.

(184) However, the disclosure is not limited to the above-described examples, and the guide device **40** may operate in various methods to guide the window covering **6**.

(185) According to one aspect of the disclosure, the air conditioner can be provided with an improved appearance quality.

(186) According to one aspect of the disclosed invention, the air conditioner can prevent degradation in cooling and heating efficiency.

(187) According to one aspect of the disclosed invention, the air conditioner can prevent window covering from covering a suction port by guiding the window covering.

(188) Although certain illustrative embodiments and implementations have been described herein, other embodiments and modifications will be apparent from this description. Accordingly, the inventive concepts are not limited to such embodiments, but rather to the broader scope of the appended claims and various obvious modifications and equivalent arrangements as would be apparent to a person of ordinary skill in the art.

Claims

1. An air conditioner comprising: a housing including: a suction port to suck air from a space adjacent to the housing into the housing, and a discharge port through which the suctioned air is discharged; a heat exchanger to exchange heat with the suctioned air through the suction port; a fan to allow the air to be suctioned through the suction port or to allow the suctioned air to be discharged through the discharge port; and a guide panel coupled to the housing and configured to be movable forward and backward between: a first position in which the guide panel is withdrawn to the housing without covering the suction port, and a second position in which the guide panel is moved linearly away from the housing at the suction port so as to prevent an object from covering the suction port.
 2. The air conditioner of claim 1, wherein the guide panel is in the first position while the air conditioner is not in operation and the guide panel is in the second position while the air conditioner is in operation.
 3. The air conditioner of claim 1, wherein the housing is installable between a ceiling inner wall and an indoor-side ceiling spaced downward from the ceiling inner wall, the air conditioner further comprising a discharge panel installable on the indoor-side ceiling to correspond to the discharge port, the discharge panel including a blade to open and close the discharge port.
 4. The air conditioner of claim 3, wherein the discharge panel is exposed downward of the indoor-side ceiling, and the housing is covered by the indoor-side ceiling.
 5. The air conditioner of claim 3, further comprising a drain cover at a lower side of the heat exchanger to collect condensed water generated in the heat exchanger, the drain cover covered by the indoor-side ceiling.
 6. The air conditioner of claim 1, wherein the guide panel is arranged in a horizontal direction such that in the first position the guide panel is accommodated in the housing, and in the second position the guide panel is horizontally moved to an outside of the housing from the first position.
 7. The air conditioner of claim 1, further comprising: a motor configured to generate power; and a power transmission member configured to transmit the power generated by the motor to the guide panel, wherein the power transmission member includes: a pinion formed to share a rotation shaft with the motor, the pinion on one of the guide panel and the motor; and a rack gear engaged with the pinion, the rack gear provided on an other one of the guide panel and the motor.
-